

INTERNSHIP TASK REPORT

ON

ESP32 PIR Motion Sensor with ThingSpeak

Task-4

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INTRODUCTION

Nowadays, technology is being used in many areas to make our lives easier and safer. One of the common uses of technology is motion detection. Motion detection systems are used in homes, offices, shops, and other places for security and monitoring purposes.

In this project, a Smart Motion Detection System is developed using a PIR sensor, ESP32, and ThingSpeak cloud platform. The PIR sensor is used to detect human movement. When a person moves near the sensor, it detects the motion and sends a signal to the ESP32.

The ESP32 receives the sensor data and connects to the internet using WiFi. It then sends the motion data to the ThingSpeak cloud platform. The data is stored in the cloud and displayed as a graph. This helps users see the motion activity in real time from anywhere with an internet connection.

The main aim of this project is to understand how IoT devices work and how data can be sent from a sensor to the cloud. This project also helps in learning about sensors, cloud storage, WiFi communication, and real-time monitoring.

This system is simple, easy to use, and can be useful for basic security and monitoring applications.

PROBLEM STATEMENT

In many homes, offices, and public places, it is important to monitor human movement for security and safety purposes. Traditional motion detection systems can detect movement, but they do not allow users to view the data remotely. There is a need for a simple system that can detect motion and send the information to the cloud for real-time monitoring. Therefore, this project aims to develop an IoT-based motion detection system using a PIR sensor, ESP32, and ThingSpeak cloud platform.

OBJECTIVE

The main objective of this project is to detect human motion using a PIR sensor and send the detected data to the ThingSpeak cloud through ESP32. The data is displayed as a real-time graph for monitoring purposes.

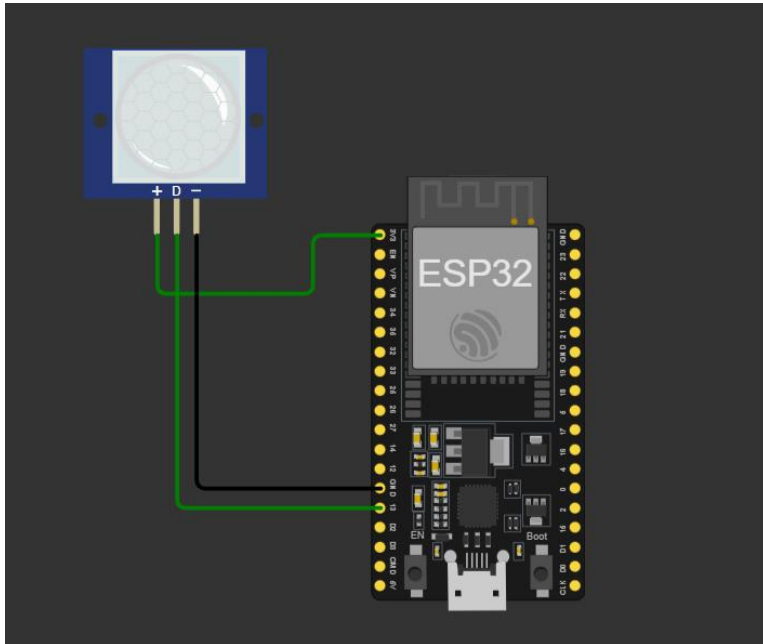
COMPONENTS REQUIRED

S.No	Component	Purpose
1	ESP32	Reads sensor data and sends it to the cloud
2	PIR Sensor	Detects human motion
3	WiFi	Connects ESP32 to the internet
4	ThingSpeak	Stores and displays data as a graph
5	Jumper Wires	Used for connections
6	Wokwi Simulator	Used to simulate the project

WORKING PRINCIPLE

The PIR sensor is used to detect human motion. When a person moves in front of the sensor, the PIR sensor detects the movement and sends a signal to the ESP32 microcontroller. The ESP32 reads the sensor value and connects to the internet through WiFi. It then sends the motion data to the ThingSpeak cloud platform using an HTTP request. The data is stored in the cloud and displayed as a graph on the ThingSpeak dashboard. If motion is detected, the value sent is 1, and if no motion is detected, the value sent is 0. This allows users to monitor motion activity in real time from anywhere using the internet.

CIRCUIT DIAGRAM



ARDUINO CODE:

```
#include <WiFi.h>

#include <HTTPClient.h>

// WiFi details

const char* ssid = "Wokwi-GUEST";

const char* password = "";

// ThingSpeak Write API Key

String apiKey = "MM7JXYQC3HUAVY17";

// PIR sensor connected to GPIO 13

int pirPin = 13;

void setup() {

    // Start Serial Monitor

    Serial.begin(115200);

    // Set PIR pin as input
```

```
pinMode(pirPin, INPUT);

// Connect to WiFi

WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {

    delay(500);

    Serial.print(".");

}

Serial.println("WiFi Connected");

}

void loop() {

    // Read motion sensor value (0 or 1)

    int motion = digitalRead(pirPin);

    // Check WiFi connection

    if (WiFi.status() == WL_CONNECTED) {

        HTTPClient http;

        // Create ThingSpeak URL with sensor value

        String url =

            "http://api.thingspeak.com/update?api_key="

            + apiKey +

            "&field1=" +

            String(motion);

        // Send data to ThingSpeak

        http.begin(url);

        // Get server response code

        int code = http.GET();
```

```
// Print response code on Serial Monitor

Serial.println(code);

http.end();

}

delay(15000);    // Wait 15 seconds before next update

}
```

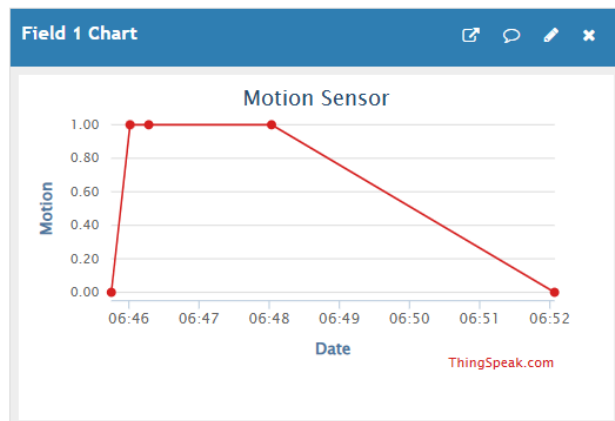
OUTPUT

Channel Stats

Created: 36 minutes ago

Last entry: 24 minutes ago

Entries: 5



APPLICATIONS

1. Home security systems to detect unauthorized movement.
2. Office and workplace monitoring.
3. Smart buildings and smart home automation.
4. Motion-based lighting systems.
5. Security monitoring in shops and warehouses.
6. Restricted area monitoring.
7. IoT-based surveillance systems.
8. Real-time motion monitoring using cloud platforms.

CONCLUSION

The IoT-Based Smart Motion Detection System was successfully implemented using a PIR sensor, ESP32, and ThingSpeak cloud platform. The PIR sensor detected motion and the ESP32 sent the data to the cloud through WiFi. The motion data was displayed as a graph on the ThingSpeak dashboard in real time. This project helped in understanding the basic concepts of IoT, cloud connectivity, and real-time monitoring. The system is simple, effective, and can be used for basic security and monitoring applications.